Sentiment and Trader Behaviour on Hyperliquid

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Github link : https://github.com/ruchishru/hyperliquid-sentiment-analysis

Methodology

- Joined 2,644 daily Fear/Greed records with 211,224 Hyperliquid trades on calendar date , found 479 overlapping trading days, no missing values .
- Mapped 10 direction values to open-long / open-short / close flags computed per-trader daily PnL, win rate, trade count, long ratio, position size, and net-of-fee PnL .
- Retained 5-class sentiment index and collapsed to binary Fear / Greed (merging Extreme variants), with Neutral as a separate bucket .
- Segmented traders into three pairs: Frequent vs Infrequent, Large vs Small Size, and Consistent Winners vs Others .
- Trained a 200-tree Random Forest (depth 8) on behavioural and sentiment features to predict daily profitability, using StandardScaler normalisation and an 80/20 split .

Strategy Recommendations

Fear Regime : Reduce Size & Frequency

- Cutting position size by ≥ 30% vs our normal baseline .
- Cap daily trade count at ≤ 50% of our Greed-day average .
- Avoiding long-biased setups without strong price confirmation .
- Staying mostly flat during Extreme Fear; favouring short if trading .

Greed Regime : Scale Up Consistent Winners

- If trailing 30-day win-rate > 50%, allow 1.5× position sizing .
- Increase in trade frequency where Greed conditions reward execution .
- Favour long-biased setups during standard Greed .
- Adding short hedges during Extreme Greed as long ratio collapses .

Using Machine Learning to Predict Profitable Trading Days

- Trained on just 6 features : trade count, position size, long ratio, volume, and sentiment signals. The Random Forest classifier hits **76% accuracy** and an **AUC-ROC of 0.81** .
- The Fear/Greed score confirms that sentiment genuinely moves the needle on profitability .
- **Profit days are called correctly 98% of the time,** so when the model says conditions look good, they usually are good .
- Loss days are harder to catch (40% recall), because bad days are less predictable by nature .